

Musings on the Brutality of Email

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Abstract

Email is a horrible thing. The following goofy but thoughtful rant describes the four Axioms of Email, analyzes our inbox using Yoblanski's *Laws of UX*, and describes exactly how and why email is so brutal. In particular, this work: identifies email as an unending background hum in the soundtrack of our lives; reveals how our inbox constantly lies to us; and explores how our brains are both awesome and terrible. Potential design solutions are suggested, and three new Laws of UX are proposed: Schrodinger's Email; the Sisyphus Effect; and Caro's Law. Readers who hate email will benefit from a better understanding of why it sucks so much; and readers who find email simple and straightforward will learn why they should stop judging the rest of us. Feel free to give a copy of this anonymously to your boss, so they're less rude next time you miss an important email (that could have just been a meeting!).

Keywords

email, rant, principles of bad design

ACM Reference Format:

Caro Williams-Pierce. 2025. Musings on the Brutality of Email. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3716233>

1 What is this even about?

I've never heard anyone say they like email. I mean, maybe they've gotten a specific email they like - *you've just won the MacArthur Genius Grant!* or *you got tenure!* or *there are cookies in the break room!* - but no-one admits to actually liking the unending background hum of an inbox filling up. Plenty of folks talk about how much they dislike email, or how they are terrible at it - but then they respond to emails within 30 seconds of receiving them! So I don't think they're talking about email with the same horror that I am - but anyway, I've been musing on the brutality of email, and now I'm sharing what I've found.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3716233>

2 Four axioms

There are four axioms I am taking as given:

- There are too many emails.
- Most emails are 'non-promotable tasks' [1] - that is, they will not advance us in our chosen career trajectory.
- The emails that *are* important - promotable tasks such as winning that Genius Grant - are mixed in with the million other *unimportant* emails; and
- You never know which it will be until you open it.

So opening an email isn't just about *reading and responding*, it's also:

- learning what the email is even about
- finding out how long it will take to respond (somewhere between 5 seconds and 4 hours, usually)
- deciding whether it should be added as a future agenda item (on an agenda document buried in Google Drive in some committee folder, if you're lucky enough to have an organized committee chair)
- seeing if whether there's a deadline that should be added to your constantly iterating Notion projects
- seeing whether it's the first of seventeen back-and-forth emails trying to schedule a meeting

and so on. In other words: you never know whether an email will be a delightful surprise, a predatory journal, a cold email by a student or a company, a complaint about grading, a grant you just got funded, a paper R&R, a perfunctory note, a dissertation you need to read by tomorrow morning, a request for a letter of recommendation from a student you remember fondly (or don't remember at all), another review request by a top journal, or an update on your tenure case. Or none of the above.¹

3 Tools to explore the digital abyss

So, we take those four axioms as background, and now, let's just think about the design of email as a communication method and user interface. I'm using Yablonski's *Laws of UX* [7], because his work incorporates various findings from psychology research. In other words, it's not just a wail: "Email is overwhelming!!!!", but rather a succinct description of original psychology research connected directly to UX/UI, with some relevant design examples. For example:

*Hick's Law: The time it takes to make a decision increases with the number and complexity of choices.*²

¹*Schrodinger's Email*: An email is simultaneously small and large, important and not important, life defining announcement and spam, until opened, at which point it coalesces into a 'funny forward' from your Aunt Kaylene who's been missing you since you left Facebook.

²For the record, Yablonski, the simplicity of the Apple TV remote does *not* equal a good design. Using that remote is like playing digital roulette - I came home the

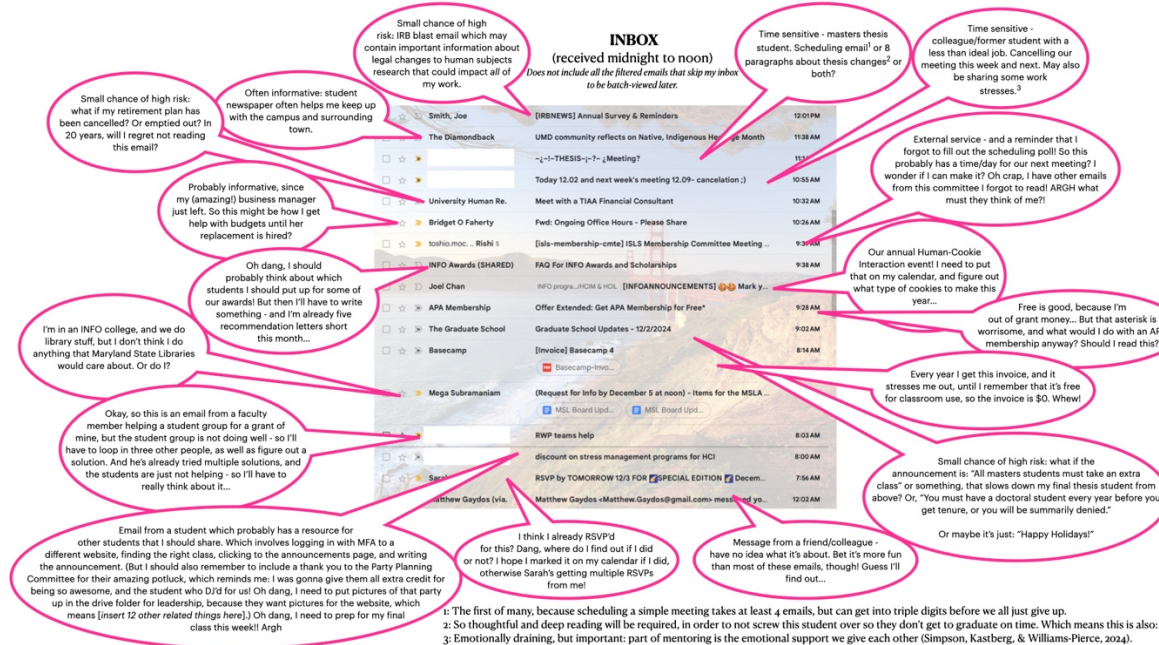


Figure 1: Some of the (potential) complexity in just a few emails. Hicks would be horrified. Some names blurred, to protect the innocent and the guilty alike.

So Yablonski's Laws of UX thread a fine line between modern digital design and hardwired human tendencies. And in the modern digital world where the last innovations in email were conversation style threading, archiving, and labels, where we accept email design paradigms even as we suffer, reaching back to before the dominant paradigm of email can be particularly revealing. All the references to laws and effects in this manuscript are from Yablonski, except *Schrodinger's Email*, the *Sisyphus Effect*, and *Caro's Law*.

4 Into the abyss

In the following sections, Yoblanski's laws of UX are used to: describe email as an unending background hum in the soundtrack of our lives; reveal how our inbox lies to us (those emails are near each other, but they are *not* actually connected!); and brains are found to be both awesome and terrible.

4.1 Why is email 'an unending background hum'?

Well, many human activities revolve around a single goal: *beating the boss* in a videogame; *writing an abstract about how awful email is*, for alt.chi; getting to work on time. Within that goal may be multiple tasks, each of which ideally advances you towards that goal. Email, however, manifests many goals and many tasks at various grain sizes (see Figure 1). And then, if we scale up a little, Inbox Zero has been an (unaccomplished) goal of mine for 20 years now. Unfortunately for my Inbox Zero goal, the *Goal-Gradient Effect* states that *the tendency to approach a goal increases with proximity to the goal*. So my lack of approaching that goal of Inbox Zero, means I am *not* increasing my tendency to approach it. I am

trapped in limbo - every step I take towards Inbox Zero, is undone by the new emails flooding in. I am getting further and further from Inbox Zero - so here's a new one for Yoblanski to add to his laws: the *Sisyphus Effect*, which states that *the tendency to move away from a goal increases with time*.

Sometimes, I sit down, take a deep breath, and think, *this is it - this is when I just work on email until it's done!!* Then, I put in a few hours of work, and guess what? The goal is no nearer. . . This is because, according to *Parkinson's Law*: *Any task will inflate until all of the available time is spent*. And, with email, the more emails you send to people, the more emails you get back! (Probably because they're thrilled you FINALLY responded!) Frankly, I feel pretty successful after spending a few hours on email and ending at net zero - the goal is to send/archive/transform into tasks/dates/events as many as I received, so even if I'm not approaching the goal, at least I'm standing still!

In other words, email is a goal that will never be achieved. And, according to the *Zeigarnik Effect*: *People remember uncompleted or interrupted tasks better than completed tasks*. So, the goal of Inbox Zero, the goal of email, is always incomplete and interrupted - and always humming away in the background of our minds.

4.2 The Lies of Proximity, Similarity, & Commonality

There are multiple 'laws' of UX that have to do with visual grouping and similarity. For example:

other day to find my Mom watching *Murder, She Wrote* when she wanted to watch the newest episode of *The Great British Bake Off*. She said she didn't mind, because the remote originally tried to make her watch *American Murder: Laci Peterson*, so at least she got to watch Jessica Fletcher.

- Law of Common Region: Elements tend to be perceived into groups if they are sharing an area with a clearly defined boundary.
- Law of Proximity: Objects that are near, or proximate to each other, tend to be grouped together.
- Law of Similarity: The human eye tends to perceive similar elements in a design as a complete picture, shape, or group, even if those elements are separated.

All of our emails look the same on the surface, presented in a common region, near each other, but can be very different (as illustrated in Figure 1). In fact, email is arbitrary – you click, and what is revealed may or may not be what you expected from the vague subject line, may or may not be a promotable task or something crucial about your retirement fund or a hard deadline for tenure. In other words, every email you open might be something that makes or breaks your career, life, family – or just another generic blast email. And there’s nothing quite as frustrating as arbitrariness. As anyone who has received arbitrary grades or played a badly designed videogame can testify, arbitrary failure is horrible, and arbitrary success is *just okay*. It’s discouraging to arbitrarily fail, because that means you don’t know what you did wrong, so you can’t learn from it [5, 6]; and it’s hard to feel successful when it was just random chance - *I finally got the Apple TV remote to actually play the show I wanted! I wonder how that happened?* - because you don’t know how it happened, or how to do it again. So you’re still forever stuck in the Apple TV remote roulette game that you never wanted to play.

The common proximity and similarity of the inbox design is what can trick our brains into believing emails belong together and are kind of the same thing, but every time we click into an email, we are dislocated into a different space and time, and different things are asked of us, with different goals, contexts, and imperatives. This means that email prevents us from accomplishing a true flow state, which – as far as I’m concerned – is the 8th Wonder of the World. Flow is a beautiful experience, where time ceases to exist, and while you might be a little tired when you re-surface, you are also excited and rejuvenated! *Flow: The mental state in which a person performing some activity is fully immersed in a feeling of energized focus, full involvement, and enjoyment in the process of the activity.*

So, if we consider the *activity* to be *answering emails*, then perhaps flow is possible for some lucky few. But I experience email as jaggedly disconnected subgoals or activities, with each new email a moment of disorientation and confusion. There is rarely a moment of full involvement or enjoyment, and I never walk away from my inbox feeling energized. Instead, I feel exhausted, like I’ve run a marathon I didn’t train for, after two weeks on a kale smoothie and apple cider vinegar diet, in a tight skirt that didn’t let me take full strides, singing the Alphabet Song backwards.

In other words, email is a LOT of work that doesn’t let you settle in and focus, but it’s disguised as similar and not much work, in part because of the above laws and also because the layout adheres to the *Law of Uniform Connectedness: Elements that are visually connected are perceived as more related than elements with no connection*. But emails only *look* connected, sitting primly next to each other, which means that the only uniform connectedness is *us* and the meaning

we lend to each email. This meaning comes from our memories of past experiences, the various platforms upon which we maintain digital traces of past emails, meetings, contexts, plans, goals, and – at least in my case – the chaotic mess of physical materials piled up in my office. So each email asks us to make these often complex connections, but frankly, our brains are not particularly good at quickly navigating the connections. Why, you ask? Because:

4.3 Our Brains May Be Awesome, But They Also Kinda Suck Sometimes

There are multiple parts to this one, so let’s start with Selective Attention: The process of focusing our attention only to a subset of the stimuli in the environment – usually those related to our goals. Email requires multiple layers of selective attention: first, actually getting a moment to sit down in front of our computer and being able to focus solely on our inbox. Then, we are handed the grab-bag of chaos that is our inbox, with very few tools to help us actually focus on a single subset – for example, I have a million student emails in my inbox, and there’s no button I can click to say “Only show me the emails from students in my Fundamentals of HCI course.” Or, “Only show me the emails asking for letters of recommendation.” Or, following some of the categories from Figure 1 “Only show me time-sensitive emails” or “Only show me the small chance of high risk emails so I can skim them and make sure my life isn’t ruined through ignorance.” And of course, since I have become a Gmail super-user in my desperate attempts to reach the promised land of Inbox Zero, I have various filters set up to label and archive emails so I can just – for example – read all the Human Computer Interaction Lab emails all together in the same folder.³ This is sometimes awesome but often faulty, and with a constant small chance of high risk – if UMD President Pines ever emails me personally, he’ll get automatically filtered into a folder of ‘random emails from UMD’, and I’ll probably never read it.⁴ Using Gmail’s default ‘category inbox’ does something similar, putting ‘social’ emails in one tab, ‘subscriptions’ in another, etc. – but I’m fairly certain that the only reason people love this inbox setting is because it automatically removes about two-thirds of your email from your inbox, so your selective attention doesn’t have to work as hard. But, of course, this just means that two-thirds of your email is arbitrarily removed from your life – so let’s just hope it’s not important, high risk, time sensitive, or from President Pines!⁵

Okay, so we’ve found a quiet space where we can focus on checking our email, and we’ve conducted a magical search that doesn’t exist anywhere but in this future dream of what email could be, and that search has returned only the emails based on one thing. This magical search supports us immensely in our selective attention – we don’t see the other emails drifting in, or keep eying the ‘small chance of high risk’ emails nervously, or let our worries

³And since LinkedIn won’t unsubscribe me from any of their 37 notification services, and Gmail refuses to treat LinkedIn emails as spam no matter how often I mark them, a filter that automatically deletes those emails is the only reason I still have some tattered shreds of patience left.

⁴This is okay, because I’ll never be that famous.

⁵And yes, I know, it’s not *actually* arbitrary – there’s an opaque algorithm that lives in a black box with an LLM or whatever. But I’m pretty sure that even the engineers most closely involved with these secrets are regularly surprised by what the system does – so yeah, close enough to arbitrary to be indistinguishable.

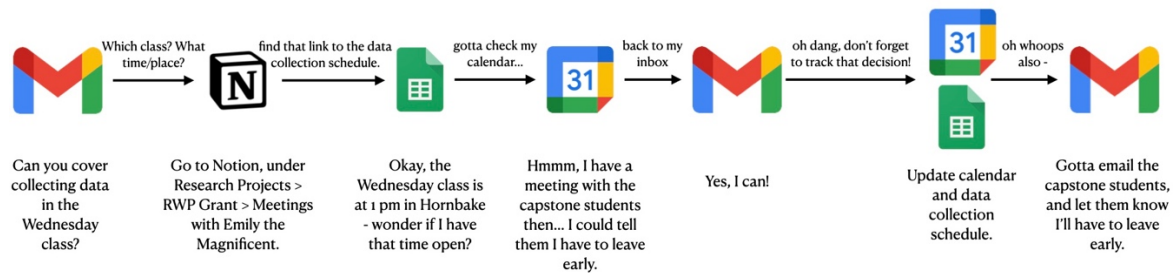


Figure 2: A Tale of One Email (and the subsequent interfaces, decisions, and connections it spawns). All icons courtesy of Wikimedia Commons.

for the thesis student interfere with the work we're trying to focus on.

In this case, let's say the 'thing' – what we're trying to focus on – is my Read, Watch, Play (RWP) grant, where we have students building a teaching and learning management system.⁶ There are numerous co-PIs across campus, four student groups working on the design in capstone classes with different faculty, many other faculty whose classes are sites of contextual inquiry by the student groups, about 700+ students who we just finished collecting research data from, an upcoming meeting with the funders who want to see pictures of the student groups doing fun-looking things, et cetera, et cetera. So there are many moving parts, and each email connects to one or more of those parts, which are tracked in different places with different people involved. This becomes a problem for our *Cognitive Load*: *The amount of mental resources needed to understand and interact with an interface* – because, of course, we're attempting to connect the RWP emails to the various online notes and resources. Despite the apparent similarity, commonality, etc. displayed by our inbox design, each email fits into a much broader digital ecosystem, with each interface and connection adding its own burden to our cognitive load. Here's a peek into part of the RWP ecosystem, following the connections forced in response to a single simple email asking me if I can take over data collection for one of the classes (Figure 2).

Just *looking* at Figure 2 makes me feel tired, but let's keep going. Okay, so we've used a nonexistent tool that let us direct our attention just to RWP-related emails, and even though those emails looked similar in the inbox, a 'simple' question still requires multi-step and multi-interface navigation. Each step and new interface add their own burden to our cognitive load, and we're not even accounting for how we go from one step/interface to the next! Because those connections – the text above the arrows in Figure 2 – also take effort, because they involve *Working Memory*: *A cognitive system that temporarily holds and manipulates information needed to complete tasks*. My working memory is what helped me remember each step – where's the link to the data collection schedule? Oh right, it's in the minutes with my meetings with Emily, which is in Notion – and so on. Working memory is what ties every little piece together, so that I am able to actually respond to that email with

something other than "Maybe?" And the last two connections – that begin with 'oh dang' and 'oh whoops' – are the dying gasps of my (admittedly neuro-atypical) working memory, trying to remind me that there's still multiple other steps/interfaces before I can go on the *next* email adventure.

5 Conclusion

In conclusion, email is terrible.

Luckily, we have one more law to help us out: *Occam's Razor*: *Among competing hypotheses that predict equally well, the one with the fewest assumptions should be selected*. Yoblanski suggests the following corollary: "The best method for reducing complexity is to avoid it in the first place." In other words, the easiest, simplest, least stressful way to manage the brutality of email is to follow *Caro's Law*: *just never check your inbox*.

6 Post-script: design recommendations

Some of my excellent reviewers wanted recommendations for fixing "the digital dumpster fire that is email," as Reviewer Yuri Kenney so perfectly wrote. So while I personally consider *Caro's Law* to be the perfect solution, here are some ideas that are less likely to get you immediately fired.

6.1 What Can I Do, As Someone Over-Blessed with an Abundance of Email?

Very little.

6.2 No, Really, Help Me!

Okay, I'll share some thoughts, but see my above point about still seeking Inbox Zero after 20 years in academia.

First, get a full-time administrative assistant to handle all your correspondence and digital nonsense. In the middle of a brilliant diatribe that I wish I could include in full here, Reviewer Michael Correll wrote, "The way to get to inbox zero last century was to make somebody else do all the thankless and annoying sorting." In other words, go find that 'somebody else' who will also save you the 2 hours it took you to book your flight to CHI using the specialized travel software your university forces you to use, that was apparently built by someone with the design goal of preventing you from spending your grant money.

⁶The term TLMS refers to a platform that helps teachers become better teachers, just as it helps students become better students. Check back in the fall, when we start using the prototype in our classes!

But then, you'll have to spend a lot of time fighting your working-class guilt for having an underpaid minion to do the work you don't want to do.

And in the midst of that guilt-fest, you might find yourself wishing you were a rich white guy professor a hundred years ago, when unabashed exploitation was considered a feature, not a bug.

Anyway, you're never going to get that assistant, so: Second, stop reading articles titled "Top 10 Tips for Productive Email" or whatever. They're just filled with idiosyncratic workarounds the author has painstakingly developed to feel like they are actually *doing something* when they check their email. But there's no silver bullet here – you'll still have to craft your own idiosyncratic workarounds that fit your life so that email will allow you to actually, finally, do your work. (And then *you'll* want to write an article sharing your top ten tips too, because you spent a lot of time figuring it all out! There's gotta be some way to make all this work into a CV line item, right?)

And look, I understand, I had the same urge when the reviewers asked for design recommendations – I started writing about filters and labels and batch-checking listservs and all the other eight hundred tricks I've developed over the past 20 years. But here's the problem: when technology doesn't work for us, we've been trained to believe that it's 'user error' – that it's our fault for not 'doing the research' to figure out how to: stop our phone from ringing with every spam call; end our Amazon Prime submission; stop our smart car from stealing all our data; or mute the TV when the remote doesn't have a mute button (yes, Apple, I'm looking at you again). Technology companies have somehow convinced us that it's *our* fault – we're just doing it wrong! But nope, we're not. Things like our inboxes are just poorly designed, and the only reason they're *still* poorly designed is because we keep making workarounds instead of just quitting our inbox!

So that leads me to the real design recommendations – the ones that require companies to actually revisit what email is, and what we – as perfectly smart, hardworking, capable, non-user-error users – need. In other words, it's time to:

6.3 Fight for the Customization We Were Promised

One promise of technology, back around the turn of the century, was individualized customization. "You'll be able to design your own stuff," we were told. "It'll be amazing!" they said. "Quality products built just for you," they said. "And cheap," they added, "because you'll only pay for the features you need," lying through their teeth. Instead, there are approximately twelve subscription tiers with every product⁷, and you're forced to look at all the features included only in the highest tier even though you can only afford Tier 2, which comes with ads every five minutes about how amazing

⁷Tier 1 (free): you can look at the product, but no touching!
Tier 2 (\$4.99/month): you can use three of the 100 functionalities of the product.
Tier 3 (\$9.99/month): Everything included in the first two tiers, plus the new and exciting ability to save your work! (Once an hour for free, additional 25 cents for each additional save per hour.)
Tier 4 (\$15.99/month): Ooooh, now you're allowed to EXPORT your work from the proprietary system!
Tier 5 (\$24.99/month): Now with no ads! (But we'll still sell your data – Tier 10 is where privacy finally kicks in.), (Hmmm, maybe a rant about subscription tiers is my next alt.chi submission.)

the tool you're using would be if you weren't too poor to pay the ransom.

For email, our institutions subscribe to the super fancy expensive tiers that brag about customization but really they're just saying, "Oh hey, you can put your logo on the inbox page! Isn't that fancy? And worth the three bazillion dollars we charge you every month? Let me upload it for you, so you can see how amazing the logo looks – see, in the top right corner, it says University of Maryland! How cool is that?!"

So, true sea changes in email design require that we get our institutions to bug the big companies who have settled on their laurels and decided to leave email as it is – after all, why fix something that ain't broke is still bringing in the money? But we have to be careful – because the only thing worse than bad design is *unfamiliar* bad design. Well, badly distributed design is pretty terrible too – instead of a single email inbox with a thousand emails, now I've got Slack chat, text messages, Discord, the oldie but goodie of text messaging, etc. etc., *and* an email inbox (still with a thousand emails, somehow?). As Reviewer Kaylea Champion wrote: "The trouble with this particular boss [of email] is that the moment you slay it, it splinters into fifty chat platforms...." Pro-tip: *Caro's Law* works for all communication systems, not just email!

But if you don't want an unfamiliar bad design, or fifty chat platforms, email has some unique design issues to consider. Despite what our inbox tells us through their proximity and visual similarity, each email has very different needs, connections to chronological time, connections to other digital and physical tools, legal implications, and so on. Redesigning around some of those main differences is the lowest-hanging fruit, especially for lives managed by single digital suites licensed through Google or a different mega-company. I'm going to tell three different 'life of an email' stories below to illustrate three points: 1) sometimes we email only because it's the easiest way to go, and a slight improvement in design could make all the difference; 2) a single small email can feel like a tiny investment in time and effort, but we're dying from a thousand cuts here; and 3) email communication can feel quick and fleeting, but email is *also*, at the exact same time, incredibly high stakes, a permanent record, and legally binding.

Finally, to make up for how mean I've been to Apple already, time to pick on Google!

6.3.1 "I'm running late to our meeting!" emails. This first 'life of an email' story doesn't begin with an email – it begins with a common situation: a calendar event, and someone is running late and wants to let everyone else know (Figure 3). In this example, I'm going to pretend that Andy is the one late to lunch at my office – so of course he reaches for his email because it's a quick and familiar way to message on a mobile device, even if he's in the middle of jogging across campus. I, the target of his email, am working on my alt.chi video submission, according to my calendar. Knowing me, and seeing the number of exclamation points,⁸ I will be so deeply focused on making the video that I will have no idea what time it is, and I *definitely* won't be checking my email. The only purpose of Andy's email is to make sure that if I *notice* his absence, I can be reassured that he's on his way.

⁸Terry Pratchett once wrote: "Multiple exclamation marks . . . are a sure sign of a diseased mind" [3]. Please don't judge me.

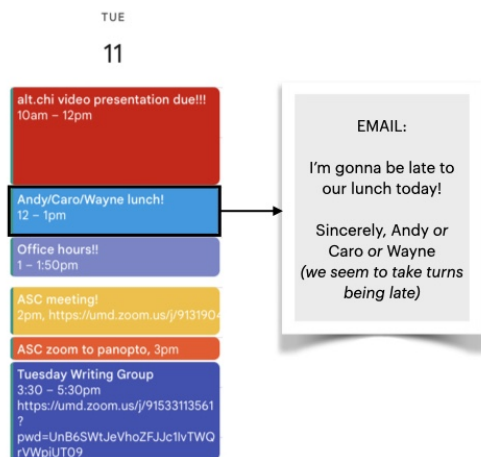


Figure 3: Screenshot of my calendar on March 11, and the lunch that one of us will probably be late for.

Chances are, however, that I won't even know what time it is until he jogs right into my office, apologizing for being late. And I'm certainly not going to check my email then – Andy's right in front of me, so the only reason to check my email (his absence) is negated. In fact, the entire point of Andy's email has fully *expired* – it is officially past its 'use by' date. But we're left with two problems with this system: 1) he actually delayed his arrival to type the email, making our precious time together even shorter, and 2) there's now a useless dead email hiding amidst a bunch of potentially important emails in my inbox. To fix #2, there is definitely a place in our modern email life for time-based archiving, especially since Google certainly knows how to parse an email that is composed entirely of "I'm going to be 20 minutes late!" sent hurriedly at 11:55 am, that can be connected to the shared calendar event starting at noon. But let's ignore the complexity of emails that 'expire' (archive) automatically, because that still doesn't account for #1: Andy having to delay his arrival further to let me know he'd be arriving late.

Instead, here's a simple two-click design change to the calendar event. A long-press on the event opens it on mobile, and there are already 1-click options to say yes/no/maybe to the event – so when it's almost time for the event, why not just add a 1-click LATE button? After all, if you're digging around in a calendar event on your mobile right before the event starts, you're probably not finally RSVP-ing for it – you're either looking for the location or for an "I'm running late to our meeting!" button.

But let's say that I do somehow return pre-Andy from the far-away land of planning my video, and remember that it's Tuesday, and think, "don't I normally eat lunch with Andy about now?" The first move I'm going to make is to my calendar – after all, maybe it was cancelled this week? Or maybe he couldn't make it so he RSVP'd no? Or maybe today is actually Wednesday? When I glance at my calendar after Andy does his long-press 1-click LATE, I immediately see the alarm clock symbol (Figure 4), and know that lunch is still happening, but will be starting a bit late. Then I can

dive back into my video work – or go answer some emails that have nothing to do with Andy telling me he's running late. In total, we've changed from multiple actions/typing by Andy that delays our meeting and clogs my inbox without accomplishing anything, to two quick actions by Andy that probably still won't accomplish anything (but at least leaves my inbox alone).

Given that I'm recommending the addition of a symbol/feature/interaction, let me be clear: Yes, I know that feature bloat is an issue! But that is the joy of good customization – the customization we were promised 20-odd years ago, but have yet to receive. Customizing isn't about 'giving everyone the same features,' it's about 'giving a single user the features that are useful to them.' If I had another 20 pages of space here, I'd wax eloquent about a bunch of different ways to do that – but let me just briefly say:

UMD, just like every institution and community, can decide which features are actually relevant to us, its users – this is not the typical feature bloat that suddenly means *every. single. user.* needs to figure out how to do a new thing on their own. After all, maybe there are companies or universities where no-one ever shows up late, so this isn't a feature worth implementing or surfacing to users! Highly unlikely, but possible, I guess? Anyway, the point I'm making with this story is that not all email should *stay* in our inbox. Instead, we should revisit types-of-email-that-should-not-be-email, and look for already familiar commonly-used places in our interwoven digital suites where those types of emails smoothly could (should) be. In this case, the calendar: it's easy for Andy to indicate he's running late, easy for me to notice it in a naturally contextual fashion (e.g., when I check the event to see if he RSVP'd or not), easy for me to ignore it if I don't need it, and where it will expire naturally by being located in the 'past' as time goes by.

6.3.2 "Parking rules are changing temporarily" blast emails, and their ilk. You've probably had this blast email experience if you live in a car-obsessed society on a parking-constrained campus with a big fancy sports team that regularly draw thousands of people. Essentially, everyone on campus gets an email warning that cars parked with some types of passes (but not all) in Lots X, Y, and Z, on such-and-such a day, at such-and-such a time, will be towed. If you don't park in that lot, or not with that pass, or not on that day or during that time, the email is irrelevant to you – but it contains so much information that you'll have to read it closely, just in case. (And if you bike to work every day, I'm sorry to tell you: repeatedly marking them as 'spam' doesn't seem to be working.)

But even if you don't work on a campus that regularly re-sells your parking spot to someone else, there are plenty of institutional blast emails flowing in just about everywhere, about everything, all the time. The current design paradigm rests on the knowledge that everyone bemoaning the state of their inbox blames themselves for: being a slow reader or writer or thinker, or a poor multitasker, or someone who just *doesn't know how to use email properly*. The only tools for simplifying the chaos are poor ones that haven't been improved in a while, *and* require that you spend forever learning how to use them. In other words, if you're not a super-user, we'll just blame user error; if you *are* a super-user, then we'll just blame you.

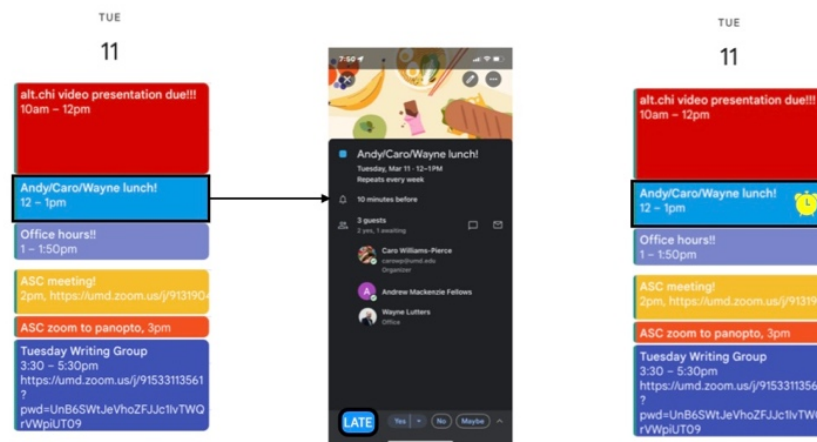


Figure 4: Andy does a long-press to open the event for the LATE button (left and middle); What I see if I glance at our shared event (right).

Instead of continuing to throw blame in the direction of us, the hapless users, let me tell you a story. A beautiful story, where email blasts are not about *sending emails*, but about *curating information*. Imagine, if you can, a world in which I only receive blast emails that actually *apply* to me! All the detailed emails that land in my inbox to be read, are only emails that I actually *do* need to read – all the other emails about parking lots I don’t park in just hit an institutional filter that strips my email out of the receiver list (see Figure 5).

And if it’s just too difficult for the multi-billion dollar company of Google to implement a system that can talk reliably to various campus databases, they could offer what I think of as ‘super-labels’: instead of all 40,000 UMD campus members individually creating their own filter that automatically labels parking-based blast emails, one person in the parking office could set a super-label that then shows up in all 40,000 inboxes. Maybe even – gasp! – super-labels could be curated well enough that I can triage at a glance whether I need to read the long detailed email, or just archive it unopened.

Super-labels are a simple enough tool that they could be locally adapted in a variety of ways – maybe a nice little ‘predicted time to read’ summary, so every official email would come with a ballpark figure for how much time it might take,⁹ or how time-sensitive it is. Urgent announcements like “Burst pipe blocking the main road into campus!” could be marked as such (and maybe even archived automatically when the pipe has been fixed and the email information has expired?). Other super-labels could be used to mark some of the different characteristics shown in Figure 1 – the email about meeting with my retirement fund managers, for example, was just an educational opportunity, not a warning that the entire fund is collapsing. A super-label that said “learning opportunity” would have helped keep my blood pressure down, just as a super-label that said “high risk” or “doom incoming” would have resulted in me reading it sooner (although along with increasing my blood pressure pretty severely).

⁹To read, anyway – “Will you send me your tenure packet?” takes about 2 seconds to read, but about 6 years to respond to.

And maybe there are super-labels that everyone can apply to the emails they’re sending – these would have to be carefully designed, sure, but here are some that I want, mostly seriously:

- The paper you’re supposed to read before we meet this week is attached
- A request to schedule a meeting
- You forgot to send me that paper about Nintendo’s Seal of Approval you promised
- I’m cc’ing you just because this jerk won’t bother responding to anyone less important than a dean
- Horrible life event means I’m going to miss class this week
- Don’t forget to sign this digital form, or UMD will keep that \$1,240 you paid out of pocket for your students
- Recommendation letter due tomorrow – did you forget me, Dr. Caro?! PLEASE READ THIS RIGHT NOW

Before I wrap up this section, I’m sure there are people who are thinking, well, sure, but it’s not like this is very labor-intensive – it’s just, what, 30 seconds to skim it and see if it applies to you?

First, you haven’t seen the parking emails that UMD sends out – they are LONG AND DETAILED and potentially high stakes, if your car gets towed.

Second, let’s say that you’re right – they’re very quick to read, and it takes just 30 seconds to:

- ⇒ Notice in Inbox
- ⇒ Open
- ⇒ Read the multi-paragraph detailed complexity
- ⇒ Decide What To Do
- ⇒ Delete Forever and Never Think About Again

Just 30 seconds, but the UMD campus population – the number of people receiving this email – is about 40,000 people. 40,000 people taking 30 seconds on this email is 20,000 minutes. That’s 165 hours – roughly 6 days and 21 hours of time being used up by this single email. What could *you* accomplish in that much time?

In other words, no matter how much UMD is paying Google for the privilege of sending these emails, it is *nothing* compared to how

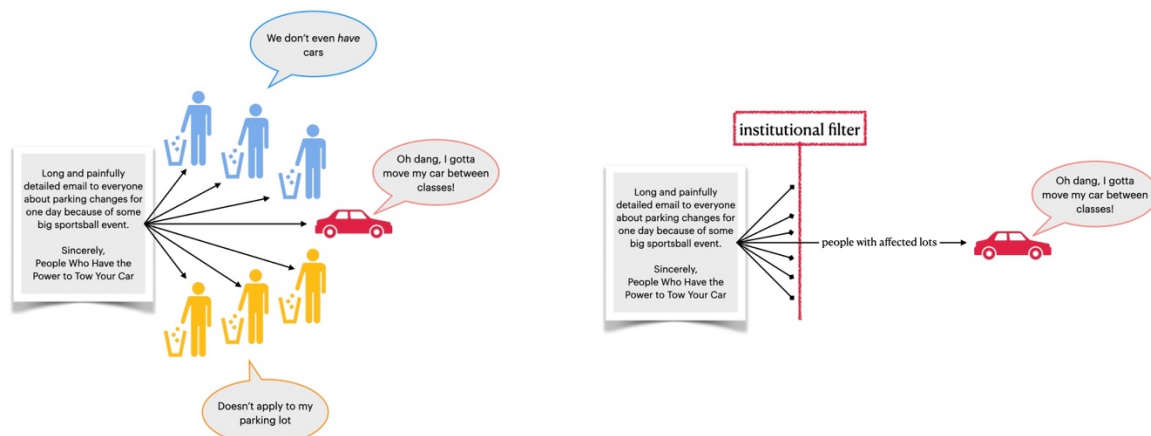


Figure 5: How current email blasts work (left); and a utopic vision of how email blasts *could* work (right).

much we, as an institution, are paying for the privilege of receiving emails that rarely apply to us. We – the users and the institution – are spending massive amounts of time just to use this antiquated system that prevents even *basic* institutional email customization.

Maybe Google should be paying us instead?

6.3.3 Legally Binding Permanent Record. Sometimes, in search of that Inbox Zero life goal, I visually skim my inbox, check off as many as possible, hit ‘mark as read,’ and then ‘archive.’ And then I absently hope that clicking the ‘mark as read’ button is not legally binding me to have actually *read* it. If I were up in court for egregious email non-response, and someone was saying, “Ma’am, it is clear that you have intentionally refused to respond to our requests for yet more service,” I feel like I should be able to rebut, “I didn’t actually *read* your request, I just hit the button that *says* I read it.”

But it turns out, in the United States legal system at least, that just having *received* an email is legally binding in some very odd ways. I’m really hoping that there’s a line drawn somewhere, because my brain has dreamed up some really weird stuff. Like, President Pines finally sends me that personal email, but a) it’s automatically archived out of my inbox because of my filters, and b) it says something like “If you don’t respond within 24 hours, you are agreeing to work for free in perpetuity. This email is a binding contract.” A month later, I don’t get a paycheck, and start wondering why – but oops! That binding contract went straight into the infinite gloom of my inbox archive, but somehow it’s the legal equivalent of signing in blood.

Reviewer Alex Leitch has been thinking about this complexity far more than I have, and they noted: “thanks to legal standards around archiving, email serves an official role as a form of institutional memory in decision-making and institutional relationships.” They then extended *Schrodinger’s Email*: “like Schrodinger’s cat existing in multiple states, each email simultaneously serves as immediate communication and historical record.” President Pines’ email is both an email he sent *and* proof that I received it – my lack of a response is beside the point. I am bound by his intention to communicate, and my emails similarly bind others.

This binding is probably why I get a new blast email every time some law, policy, funding change, etc., influences UMD – the institution has both the resources *and* the legal responsibility to keep us informed. But it turns out that emails are possibly the worst communication method in history for this. First, emails are traditionally brief: if the predicted reading time summary on an email said “this will take 6 hours to read, and a law degree to understand,” very few people would read it. But at the same time, emails are inherently decontextualized: it’s not enough to say, “The federal government has decided to defund any research it doesn’t understand.” There’s just too much complexity in the recent changes (and un-changes, because apparently the federal government doesn’t understand how to govern either), to fit into a single email. Not to mention the many lawsuits that are moving against this federal obsession with the scientific equivalent of red-lining – daily, sometimes hourly, legal changes are coming in and going out. You might get one email saying, “stop research that involves accessibility!” and then a follow-up email, “actually, keep going, there’s an injunction against that until Tuesday.”

Okay, this may not be the best example for the point I want to make, but I’m a little cranky about all this nonsense.

The point I’m *actually* trying to make is that usually, in less halcyon days, I’ll get a few emails a week about little legal changes that I don’t understand, because the emails are brief and decontextualized. The information has to be shared in a timely fashion, because of the institution’s responsibilities to us, but the information also *can’t* be brief (because it’s legalese) and *can’t* be decontextualized (because it was shaped within a complex legal context). So what I recommend is a weekly digest-style email that briefly references each new policy or change and points directly to the specific contextualized change. Specifically, when you click on one of those brief references, an internal webpage opens with the full context of that reference – the entire law, or whatever, but you are immediately brought to the part of the reference that has been changed, with a track-changes style that makes the change easy to find, within the context of the rest of the (unchanged) law. In my dream of a better system, I also want each of these webpages to be built with a layered Wikipedia-like system, so it’s easy to see the changes that

have been happening over time, to add historical contextualization as well.

All of this goes to say: for all the serious responsibility an email carries, there are certainly a lot of them flying around willy-nilly, expecting to be read. But we can use email less, as a habitual method of notification that piques our interest¹⁰ and easily steers us to more detailed, contextualized information.

6.4 Three Things I Do About Email, Just in Case You Really Want Them

Step 1: every March, archive emails from the previous year. If you haven't answered them yet, you probably never will, so do yourself this one small kindness. I am once again anxiously awaiting March 1, when the 850 or so emails in my inbox from 2024 will finally get to disappear. And then maybe, just maybe, a solid 8 hours of email on March 1 will get me to the elusive Inbox Zero. . . (A girl can dream, right?)

Step 2: make every email a meeting. No, seriously, I'm not even kidding – I would much rather have quick meetings to hash out scheduling, questions about grading, talk about recommendation letters, etc., then do the volleying of email questions, clarifications, and Doodle polls that just illustrate what we already knew (there's no time that works for everyone).

Step 3, which is really just about how to make Step 2 possible: Have office hours every week in a place public enough that even your colleagues can drop in with questions, and include a link in your email signature to digital office hours that people can quickly book a timeslot in. Be explicitly clear to everyone you encounter that if you don't respond to their email, they can just: grab you in the hallway, chat at the end of class, take it on faith that you noticed their email about a sick grandmother, come for a walk with you and your oversized pup every morning, add you as a consultant to their grant so you *have* to show up to the weekly meetings, send vibes and hope you have ESP, slide a note under your office door, or any of the eight million other ways to communicate with you that works just as (more) efficiently as sending you an email.¹¹ Important note: if you implement Step 3, be prepared to apologize profusely every day of your working life for missing whatever email was sent by the person you're talking to. If you hate apologizing more than you hate email, this step is not for you.

7 The real AND FINAL conclusion, which includes the actual fix to all our email woes

In my spare time, instead of checking email, I've been developing what I call my 'human viz project,' that traces people and their tools as they navigate through time, a designed antidote to the many splintered systems we currently navigate through. Human viz both centers and follows the individual, instead of forcing them to maintain mental bridges across a dozen discrete digital systems. If this paper resonated with you, and if you found yourself nodding and chuckling and maybe even crying a little, then find me at CHI

¹⁰Or bores us to tears, so we archive that weekly digest filled with links to legally binding documents without reading it. Same outcome, fewer emails.

¹¹Reminder for anyone who needs to hear it: Do not text me to ask if I've seen your email. That is the best way to make me feel a combination of guilt, shame, imposter syndrome, and rage – all of which will make it even less likely I will ever respond to your email/text/carrier pigeon.

and ask me about human viz. I'll talk your ear off about A-series and B-series time [2] and how the modern obsession with A-series time has misled us in our design and use of digital organization and communication systems. Then I'll blather on about my design work with human viz, with occasional sidetracks into irritated rants about how the bad design of email and other tools is so normalized that they are considered of little consequence. Adapting to consistent difficulty and frustration is a human superpower, yes, but just because we don't notice that constant humming in the back of our minds anymore, doesn't mean that we shouldn't try to end it (or at least turn the knob down – sometimes that humming is *really loud*). The triviality of email may be a familiar weight around our necks, but that's no reason to let it keep sitting there. Human viz is how I'm moving towards that lighter, sweeter life.

In other words: want email fixed? Fund my sabbatical.

Acknowledgments

Many thanks to the reviewers for their thoughtful and truly delightful reviews – I've never had reviews that were so hilarious *and* insightful, all at once! I've also never had reviews that I so wanted published alongside my paper – it feels like there's a secret bubble of brilliance that just needs to be revealed to the world! Thankfully, four generous reviewers gave me permission to explicitly quote and name them in this manuscript. Apologies to President Pines and Andy for putting them into starring roles in my email-based fiction. Lastly, premature additional thanks to the travel grant that I hope I get, so this trip doesn't have to go on a credit card that it'll take me five years to pay off. So, many thanks to [travel support fund, TBD <fingers crossed>].

References

- [1] Babcock, L., Peyser, B., Vesterlund, L., & Weingart, L. R. 1. (2022). *The No Club: putting a stop to women's dead-end work*. First Simon & Schuster hardcover edition. New York, Simon & Schuster.
- [2] Gell, A. (1992). A-series:B-series::Gemeinschaft:Gesellschaft::Them:Us. *The anthropology of time: Culture construction of temporal maps and images*. Oxford: Berg.
- [3] Pratchett, T. (1990). *Eric: A Novel of Discworld*. Gollancz.
- [4] Simpson, A., Kastberg, S., & Williams-Pierce, C. (2024). Found poetry: Highlighting reciprocal mentor relationships and role shifting within an informal peer mentoring group. In C. Craig (Ed.), *Mentoring in STEM through a female identity lens: heroes make a difference for women* (pp. 141-166). Vernon Press.
- [5] Williams-Pierce, C. (2019). Designing for mathematical play: Failure and feedback. *Information and Learning Sciences*, 120(9/10), 589–610. <https://doi.org/10.1108/ILS-03-2019-0027>
- [6] Williams-Pierce (2021). Design is a Contract. *Sparks of Innovation: Stories from the HCIL*. <https://medium.com/hcil-at-umd/design-is-a-contract-2fd7dc07dcc5>
- [7] Yoblanski, J. (2024). <https://lawsofux.com>. Retrieved November 14, 2024. Based on his book, *Laws of UX: Using Psychology to Design Better Products & Services* (2nd edition).